Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program: CPE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed: 10/22/26**
Section: CPE11S4	**Date Submitted: 10/22/26**
Name: Gianne Mathew U. Uy	**Instructor: Engr. Noel Jason Lusung**

1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

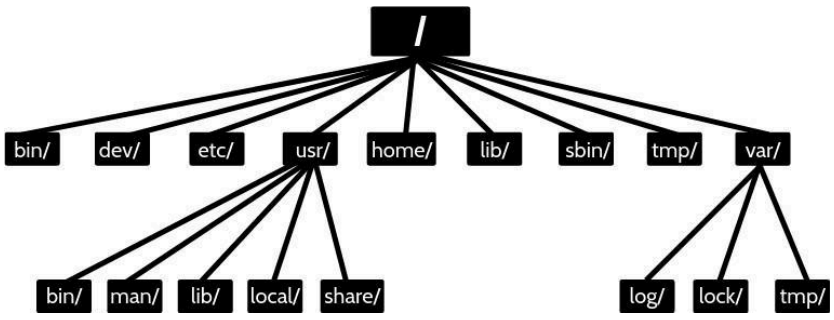
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer*.** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should

automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

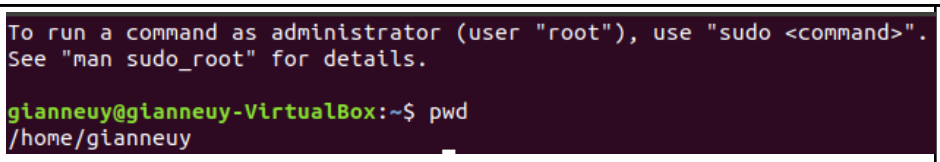
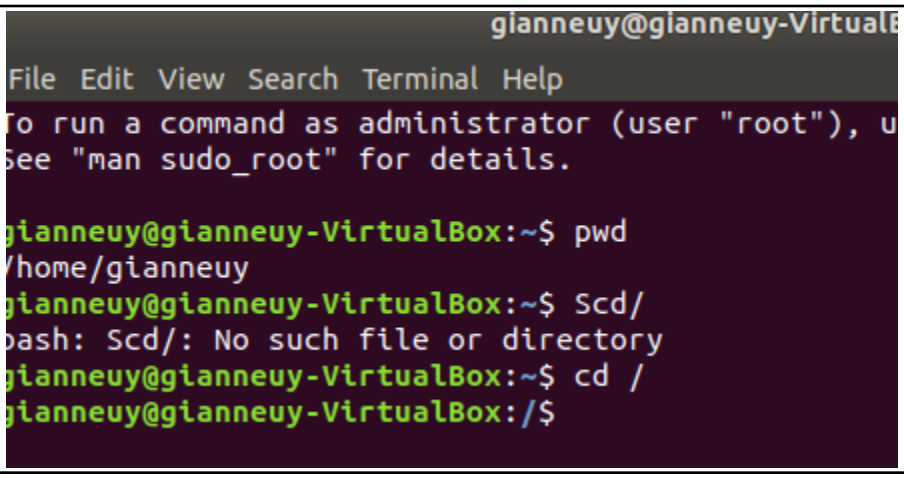
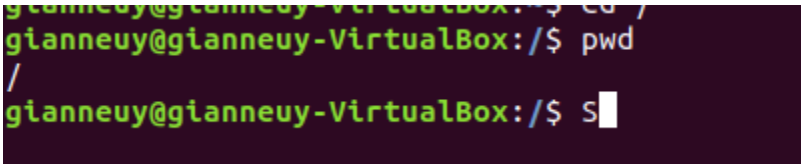
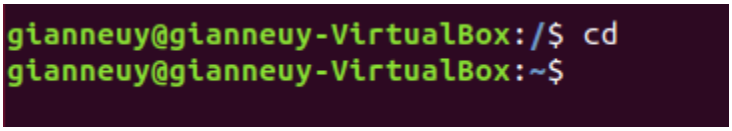
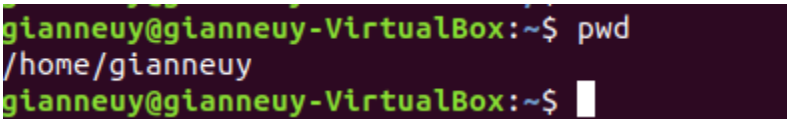
4. Resources:

Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

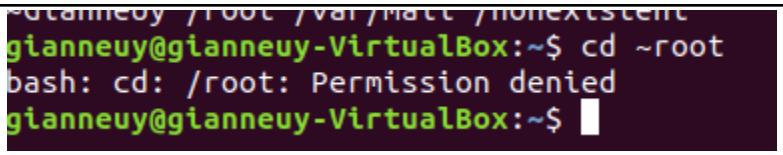
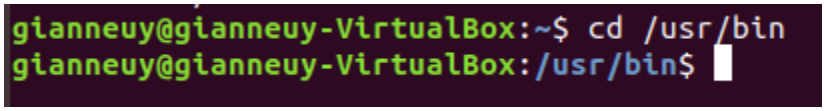
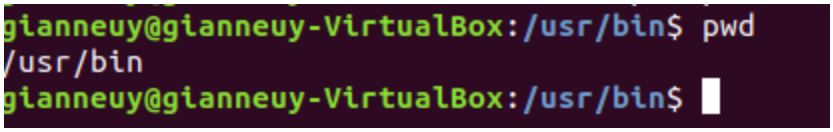
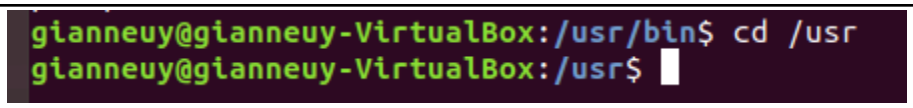
5. Procedure:

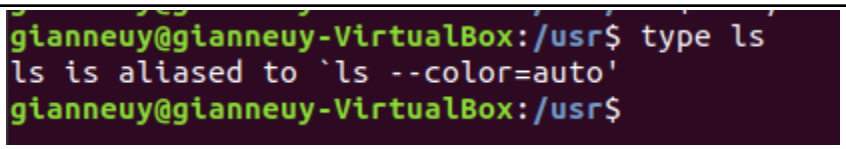
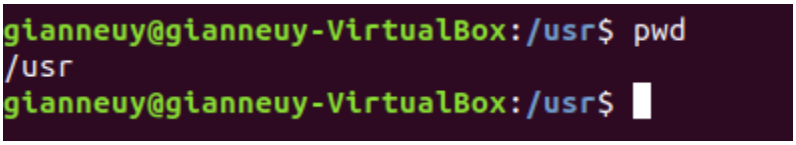
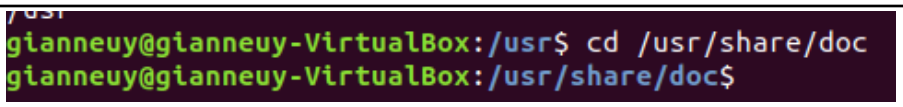
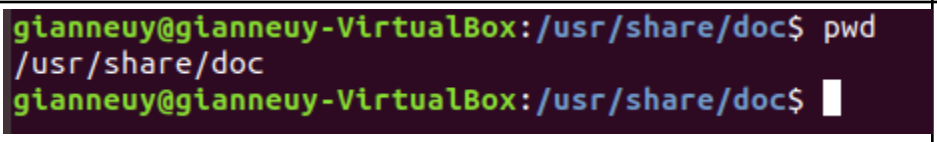
1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

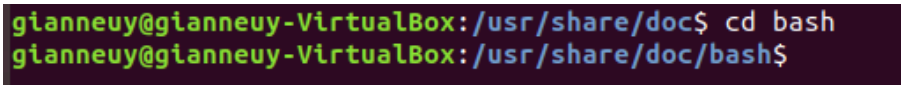
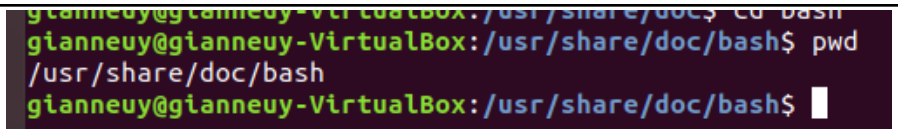
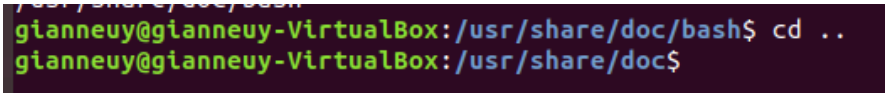
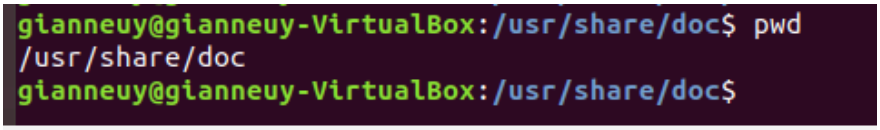
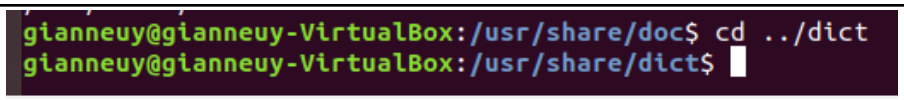
Command	Screenshot	Explanation

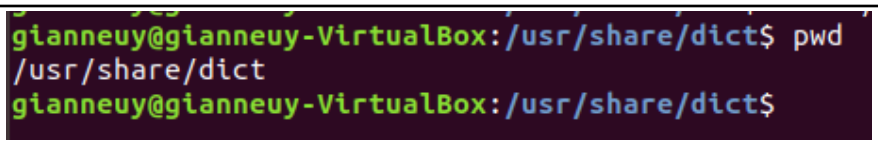
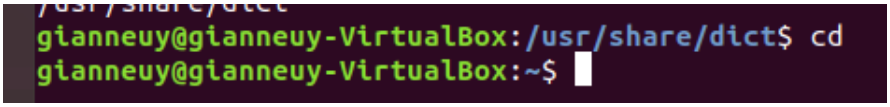
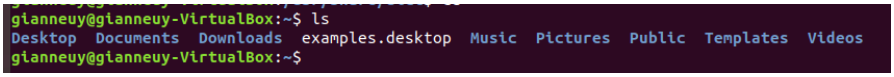
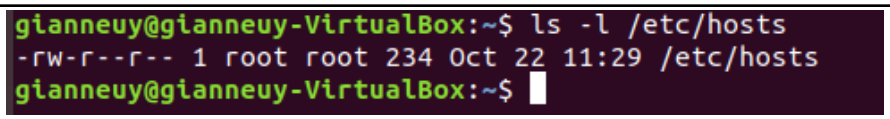
1. pwd		Displays the full path of the current working directory.
2. cd /		Changes the current directory to the root directory.
3. pwd		Displays the current directory, which should be /.
4. cd		Returns to the user's home directory.
5. pwd		Displays the full path of the current directory.

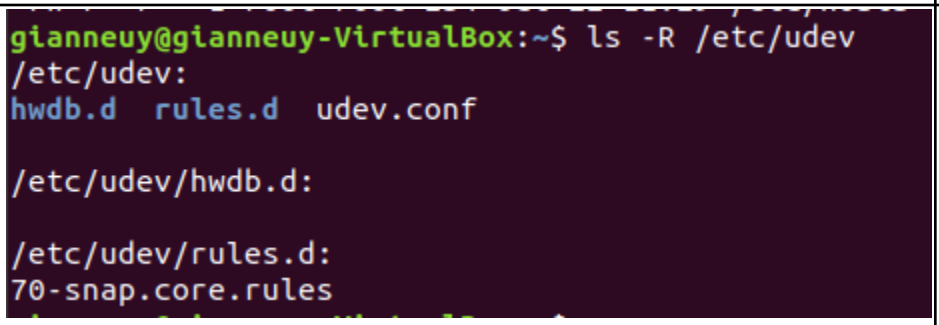
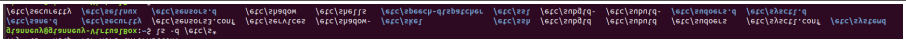
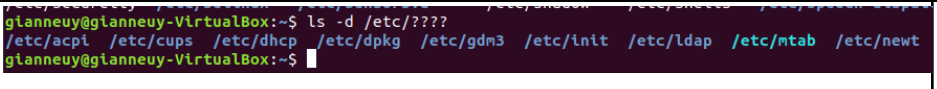
6. cd /home	/home/gianneuy gianneuy@gianneuy-VirtualBox:~$ cd /home gianneuy@gianneuy-VirtualBox:/home$	Changes the current directory to /home, where user home directories are usually located.
7. pwd	gianneuy@gianneuy-VirtualBox:/home$ pwd /home gianneuy@gianneuy-VirtualBox:/home$	Displays the current directory, which should be /home.
8. cd ~	/home gianneuy@gianneuy-VirtualBox:/home$ cd ~ gianneuy@gianneuy-VirtualBox:~$	Changes the current directory to the user's home directory.
9. pwd	gianneuy@gianneuy-VirtualBox:~$ pwd /home/gianneuy gianneuy@gianneuy-VirtualBox:~$	Displays the full path of the user's home directory.
10. echo ~ ~sysadmin ~root	gianneuy@gianneuy-VirtualBox:~$ echo ~GianneUy ~root ~mail ~nobody ~GianneUy /root /var/mail /nonexistent gianneuy@gianneuy-VirtualBox:~$	Displays the path to the user's

~mail ~nobody where sysadmin is your username		home directory.
11. cd ~root		Attempts to change to the root user's home directory. Access may be restricted.
12. cd /usr/bin		Changes the current directory to /usr/bin, which contains many executable programs.
13. pwd		Displays the current directory, which should be /usr/bin.
14. cd /usr		Changes the current director

		y to /usr.
15. type ls		Identifies how the shell interprets the ls command.
16. pwd		Displays the current working directory, which should be /usr.
17. cd /usr/share/doc		Changes the current directory to /usr/share/doc, which contains documentation for installed software.
18. pwd		Displays the current directory, which should be

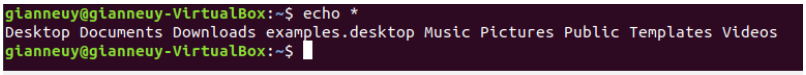
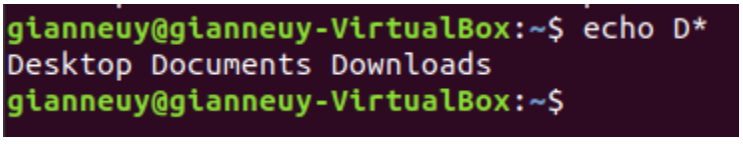
		/usr/share/doc.
19. cd bash		Changes to the bash directory inside the current directory, if it exists.
20. pwd		Displays the full path of the current directory.
21. cd ..		Moves one level up to the parent directory.
22. pwd		Displays the current directory after moving to the parent directory.
23. cd ../dict		Attempts to enter the dict directory

		located in the parent directory.
24. pwd		Displays the current directory if the previous command succeeds.
25. cd		Returns to the user's home directory.
26. ls		Lists visible files and directories in the current directory.
27. ls -a		Lists all entries, including hidden files and directories.
28. ls -l /etc/hosts		Displays detailed informa

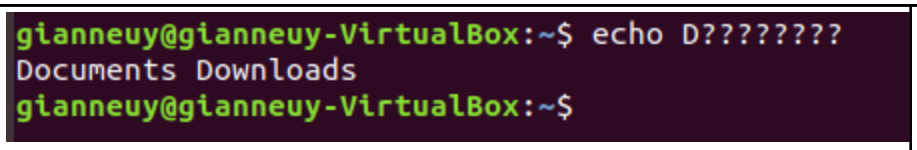
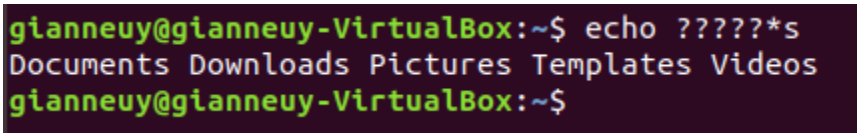
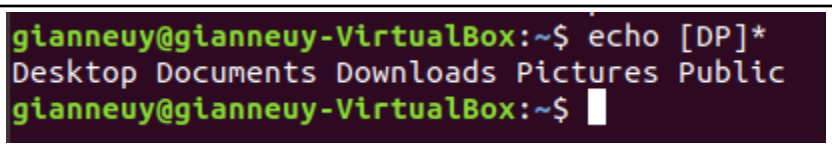
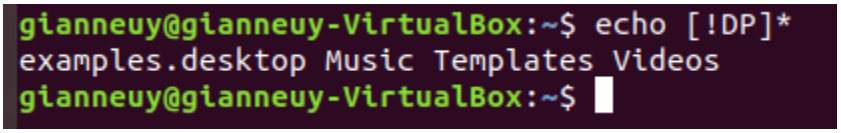
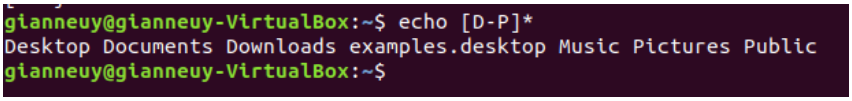
		tion about the /etc/hosts file, such as permissions, owner, size, and modification time.
29. ls -R /etc/udev		Lists the contents of /etc/udev and its subdirectories recursively.
30. ls -d /etc/s*		Displays the names of entries in /etc that match the patterns*, without listing their contents.
31. ls -d /etc/????		Displays entries in /etc

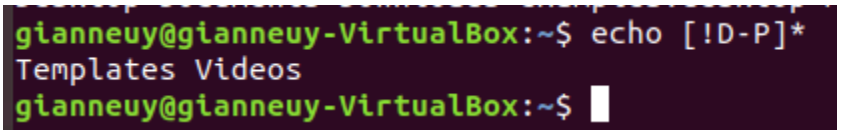
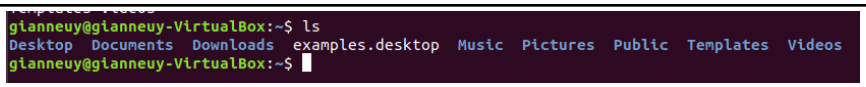
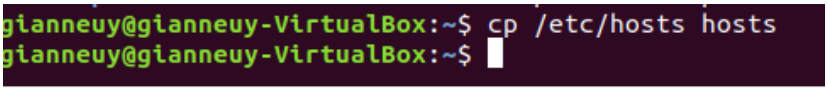
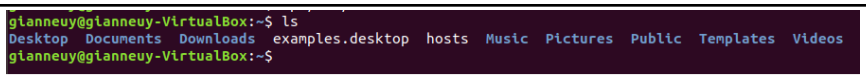
		whose names contain exactly four characters.
32. ls –d /etc/[abcd]*		Displays entries in /etc whose names begin with a, b, c, or d.

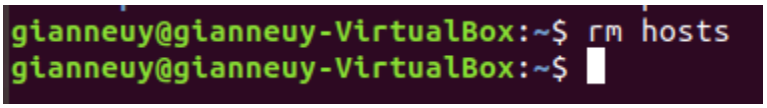
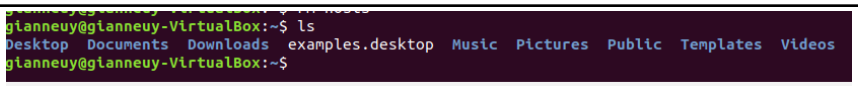
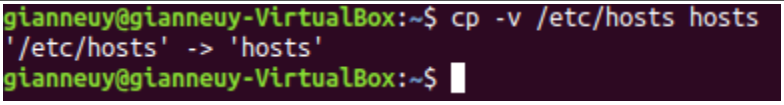
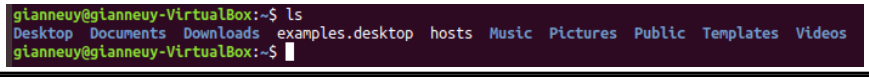
4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

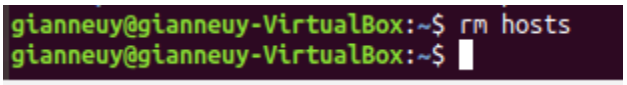
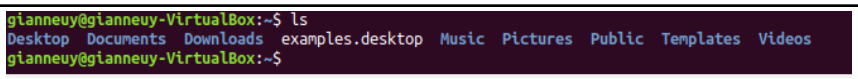
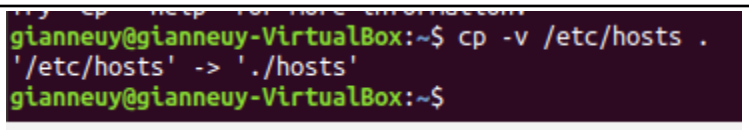
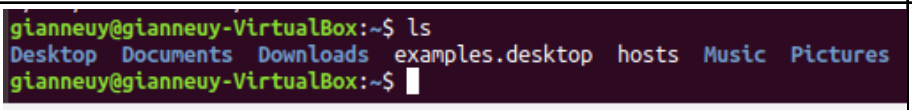
Command	Screenshot	Explanation
1. echo *		Displays the names of visible entries in the current directory that match *.
2. echo D*		Displays names that begin with the letter D.

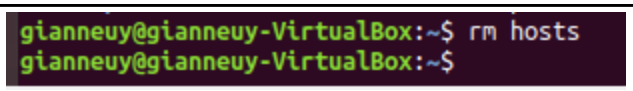
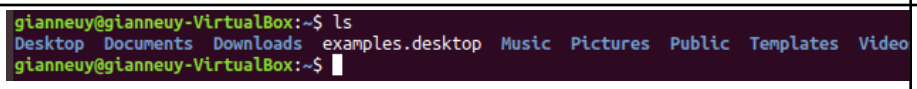
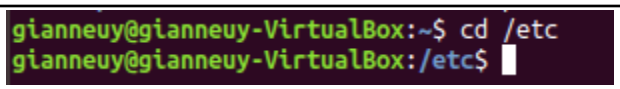
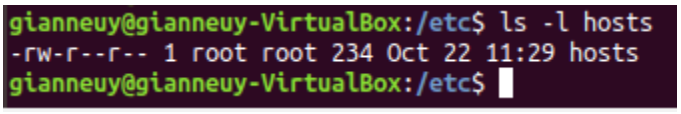
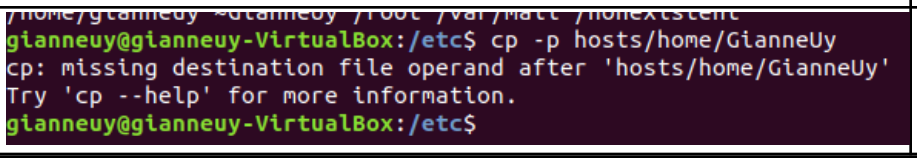
3. echo P*	gianneuy@gianneuy-VirtualBox:~$ echo P* Pictures Public gianneuy@gianneuy-VirtualBox:~$	Displays names that begin with the letter P.
4. echo *s	gianneuy@gianneuy-VirtualBox:~$ echo *s Documents Downloads Pictures Templates Videos gianneuy@gianneuy-VirtualBox:~$	Displays names that end with the letter s.
5. echo D*n*s	Documents Downloads Pictures Templates Videos gianneuy@gianneuy-VirtualBox:~$ echo D*n*s Documents Downloads gianneuy@gianneuy-VirtualBox:~$	Displays names that begin with D, contain n later in the name, and end with s.
6. echo ??????	gianneuy@gianneuy-VirtualBox:~$ echo ?????? Public Videos gianneuy@gianneuy-VirtualBox:~$	Displays names consisting of exactly six characters.

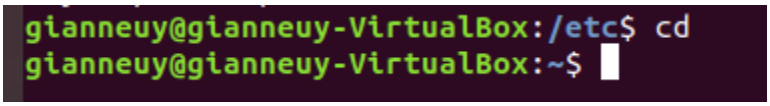
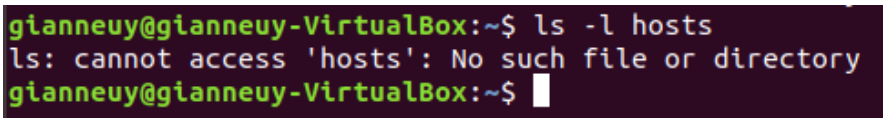
7. echo D????????		Displays names that begin with D and contain exactly nine characters.
8. echo ?????*s		Displays names with at least six characters that end with s.
9. echo [DP]*		Displays names that begin with either D or P.
10. echo [!DP]*		Displays names that do not begin with D or P.
11. echo [D-P]*		Displays names beginning with a

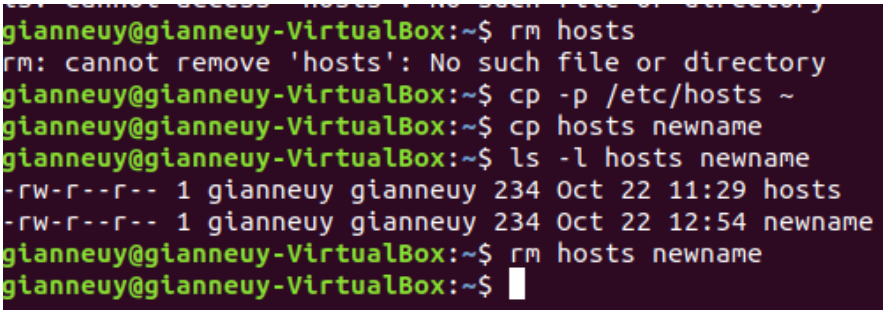
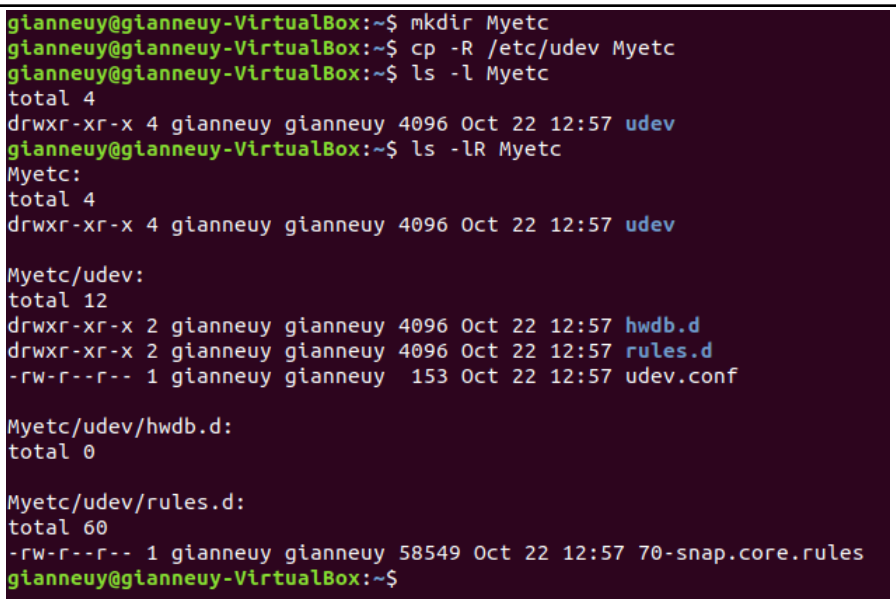
		character in the range D through P.
12. echo [!D-P]*		Displays names that do not begin with a character in the range D through P.
13. ls		Lists the visible files and directories in the current directory.
14. cp /etc/hosts hosts		Copies /etc/hosts into a file named hosts in the current directory.
15. ls		Checks the current directo

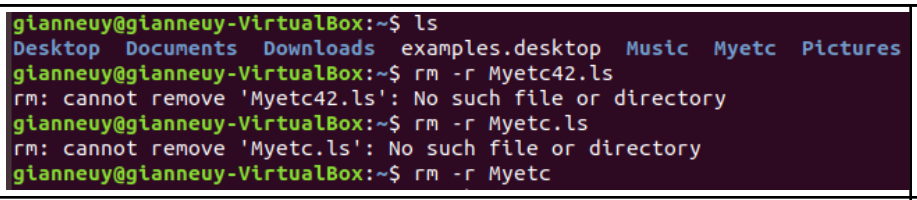
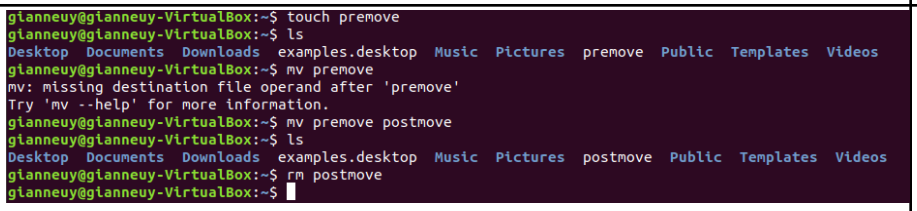
		ry to see wheth er the copied file exists.
16. rm hosts		Delete s the file named hosts from the current directo ry.
17. ls		Lists the current directo ry to verify that hosts has been remov ed.
18. cp –v /etc/hosts hosts		Copies /etc/ho sts to hosts and display s a messa ge descri bing the copy operati on.
19. ls		Lists the

		current directory to check for the copied file.
20. rm hosts		Removes the file named hosts.
21. ls		Checks the current directory after the file is deleted.
22. cp –v /etc/hosts .		Copies /etc/hosts into the current directory. The dot (.) represents the current directory.
23. ls		Lists the current directory to verify that the copied file exists.

24. rm hosts		Deletes the copied hosts file from the current directory.
25. ls		Lists the current directory after removing the file.
26. cd /etc		Changes the current directory to /etc, which contains system configuration files.
27. ls -l hosts		Displays detailed information about the hosts file in /etc.
28. cp –p hosts/home/sysadmin		Copies the hosts file to

		/home/sysadmin while preserving its original attributes, such as permissions and timestamps where permitted.
29. cd		Returns to the current user's home directory.
30. ls –l hosts		Displays detailed information about the hosts file in the home directory, if it exists.

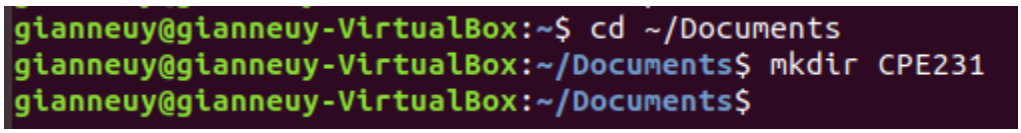
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		These commands copy the hosts file, check its details, and delete the copied files using cp, ls, and rm.
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		These commands create a directory named Myetc, copy the /etc/udev directory and its contents into it, list the copied files, and then remove Myetc.

40. ls 41. rm -r Myetc42. ls		
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		These commands create an empty file named premove, check the directory contents, rename the file to postmove using mv, and finally delete it using rm.

6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

1. Create a directory CPE231 in the Documents directory.



2. Create a directory <lastname> and <firstname> under CPE231 directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ mkdir CPE231/UY CPE231/GianneMathew
gianneuy@gianneuy-VirtualBox:~/Documents$ ls CPE231
GianneMathew  UY
gianneuy@gianneuy-VirtualBox:~/Documents$
```

3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ touch CPE231/UY/A.txt CPE231/UY/B.txt
gianneuy@gianneuy-VirtualBox:~/Documents$ ls CPE231/UY
A.txt  B.txt
gianneuy@gianneuy-VirtualBox:~/Documents$
```

4. Copy the content of <lastname> directory to <firstname> directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ cp CPE231/UY/* CPE231/GianneMathew/
gianneuy@gianneuy-VirtualBox:~/Documents$ ls CPE231/GianneMathew
A.txt  B.txt
gianneuy@gianneuy-VirtualBox:~/Documents$
```

5. Delete the content of <lastname> directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ rm CPE231/UY/*
gianneuy@gianneuy-VirtualBox:~/Documents$ ls CPE231/UY
gianneuy@gianneuy-VirtualBox:~/Documents$
```

6. Create a directory CPE231_backup in the Documents directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ mkdir CPE231_backup
gianneuy@gianneuy-VirtualBox:~/Documents$ ls
CPE231  CPE231_backup
gianneuy@gianneuy-VirtualBox:~/Documents$
```

7. Copy the content of CPE231 to CPE231_backup directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ cp -r CPE231/. CPE231_backup/
gianneuy@gianneuy-VirtualBox:~/Documents$ ls -R CPE231_backup
CPE231_backup:
GianneMathew  UY

CPE231_backup/GianneMathew:
A.txt  B.txt

CPE231_backup/UY:
gianneuy@gianneuy-VirtualBox:~/Documents$
```

8. Remove the CPE231 directory.

```
gianneuy@gianneuy-VirtualBox:~/Documents$ rm -r CPE231
gianneuy@gianneuy-VirtualBox:~/Documents$ ls
CPE231_backup
gianneuy@gianneuy-VirtualBox:~/Documents$
```

7. Conclusion:

- In conclusion, this activity helped me understand how to use basic Linux commands to manage files and directories. I learned how to navigate different folders, create and copy files, rename them, and delete files or directories when needed. At first, some commands were confusing, but

as I practiced, I became more familiar with how they work. Overall, this activity helped me improve my basic skills in using the Linux terminal, which will be useful in learning more about computer system administration.

8. Assessment (Rubric for Laboratory Performance):